Diagnose sjSDM Model Convergence

Description

Assesses whether a fitted sjSDM model has converged by analysing the training loss history stored in the model object.

Usage

```
diagnose_jsdm_convergence(mod_jsdm = NULL)
```

Arguments

`mod_jsdm` A fitted sjSDM model object. Must be of class 'sjSDM'.

Details

The function uses the 'history' component of the fitted sjSDM model, which stores the per-epoch negative log-likelihood values produced during gradient descent.

sjSDM pre-allocates 'history' to the full epoch budget. When early stopping fires, the remaining trailing entries are zero. This function detects those trailing zeros, truncates the vector to only the epochs that were actually run, and sets 'early_stopping_triggered = TRUE'.

Two diagnostic metrics are computed on the tail (final 10 epochs actually run):

1. **Linear trend slope** — A near-zero slope means the loss is no longer decreasing, indicating convergence. The recommended threshold is < 0.01. 2. **Median difference** — A robust comparison of the first versus last quarter of the tail using medians (insensitive to spikes). The recommended threshold is < 1.

Value

A list with six elements: - 'linear_trend_slope': Absolute slope of a linear trend fitted to the final 10 convergence. - 'median_diff': Absolute difference between the median of the first and last 25 convergence. - 'convergence_plot': A ggplot2 object showing the full loss history with a 20-epoch rolling median smoother. A dashed orange vertical line marks the last epoch run when early stopping was triggered. - 'note': A character string summarising the thresholds. - 'epochs_run': Integer. The number of epochs actually run before early stopping halted training (or the full budget when early stopping was not triggered). - 'early_stopping_triggered': Logical. 'TRUE' when the model stopped before exhausting its epoch budget (i.e. trailing zeros were detected in 'mod_jsdm[["history"]]').

See Also

[fit_jsdm_model()], [evaluate_jsdm()]